FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Fluids & Electrolyte Imbalances

Tutorial | Week 2 · Diarrhoea, fluids, HIV & anaemia | Paediatrics | Dr Namugga

Children need fluid prescribed like a drug, by weight (Holliday–Segar 100/50/20 mL/kg/day or 4/2/1 mL/kg/h), as an isotonic fluid with glucose, with boluses only for true shock; sodium must be corrected slowly in both directions, and potassium is given only when the child is passing urine and never as a fast IV push.

Must know before the ward round

1. Maintenance (Holliday–Segar): 100 mL/kg/day for the first 10 kg + 50 mL/kg/day for the next 10 kg + 20 mL/kg/day above 20 kg; hourly = 4/2/1 mL/kg/h. UCG 2023 uses 25 mL/kg/day above 20 kg.
2. Use isotonic maintenance fluid (Na 131–154 mmol/L, e.g. 0.9% saline with 5% glucose, or Ringer's lactate with glucose) (NICE NG29); never give plain 5% dextrose as maintenance: it causes hyponatraemia.
3. Sick children (meningitis, pneumonia, bronchiolitis, post-op) retain water (ADH): give 2/3 maintenance (NICE 50–80%) and check sodium daily. In meningitis, WHO and Cochrane (2016) favour full isotonic maintenance unless Na is low or ICP is raised (see Meningitis).
4. Shock (cold hands + CRT > 3 s + weak fast pulse): UCG 20 mL/kg RL/NS as fast as possible; WHO ETAT 2016 10–20 mL/kg over 30–60 min; reassess after each bolus. No rapid bolus if only 1–2 signs (FEAST).
5. SAM with shock: 15 mL/kg over 1 h of a glucose-containing fluid; severe anaemia with shock: blood, not more crystalloid.
6. Deficit (mL) = % dehydration × weight (kg) × 10. Add it to maintenance; most guidelines replace a deficit over 24 h (hypernatraemia over 48 h or more).
7. Neonates (UCG): day 1 60, day 2 90, day 3 120, day 4 onwards 150 mL/kg/day; 10% glucose in the first 2 days.
8. Hyponatraemia with seizures: 2 mL/kg of 2.7–3% saline over 10–15 min (max 100 mL), up to 3 doses (NICE); then raise Na by no more than 8 mmol/L/24 h (NICE max 12).
9. Hypernatraemia: lower Na by ≤ 0.5 mmol/L/h (≤ 10–12 mmol/L in 24 h); fast correction = cerebral oedema and convulsions.
10. Hypokalaemia (K < 3.5): oral KCl 1–2 mmol/kg/dose (max 20 mmol/dose) preferred; IV on the ward 0.2 mmol/kg/h (max 10 mmol/h); max 40 mmol/L in a peripheral line; only if passing urine; check magnesium.
11. Hyperkalaemia with ECG changes: 10% calcium gluconate 0.5 mL/kg IV over 5–10 min (protects the heart), then salbutamol neb 2.5 mg (≤ 25 kg) or 5 mg, insulin 0.1 unit/kg + 10% glucose 5 mL/kg; stop all potassium.
12. Weigh daily, keep a fluid balance chart, and check electrolytes and glucose before IV fluids and at least every 24 h while they run.

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1. Definitions and Key Concepts

Children are **small, wet and fast**: more of their body is water, they turn it over quickly, and they cannot regulate their own intake when they are sick. Errors of a few hundred millilitres, or the wrong type of fluid, cause **hyponatraemic seizures, cerebral oedema, heart failure or death**.

Safe fluid prescribing in children has five habits:

1. **Weigh the child** and calculate by weight.
2. **Decide why** you are giving fluid: resuscitation (bolus), deficit (dehydration), maintenance, or replacement of ongoing losses.
3. **Choose an isotonic fluid** and add **glucose** for maintenance; add **potassium** only once urine is flowing.
4. **Give less** than full maintenance to most sick children (ADH is high).
5. **Monitor**: weight, fluid balance, sodium, potassium and glucose.

> **CLINICAL PEARL**
> The one thing never to forget: **plain 5% dextrose is not a maintenance fluid.** Once the glucose is used, it is pure water. In a child with meningitis or pneumonia (high ADH), it drops the sodium and causes seizures and brain swelling. Use 0.9% saline (or Ringer's lactate) **with** 5% glucose.

Term	Definition
Maintenance fluid	Volume needed to replace normal losses (urine, stool, insensible) in a child who is not eating or drinking
Deficit	Fluid already lost (dehydration), estimated as % of body weight
Ongoing losses	Continuing diarrhoea, vomiting, NG aspirates, drains, burns, polyuria; replace volume for volume
Resuscitation (bolus)	Rapid isotonic fluid to restore circulation in **shock**
Shock (WHO ETAT 2016)	**All three**: cold extremities, CRT > 3 s, weak and fast pulse
Impaired circulation	Only 1–2 of the three shock signs: **no bolus** (FEAST)
Isotonic fluid	Sodium 131–154 mmol/L (0.9% saline, Ringer's lactate/Hartmann's, Plasma-Lyte 148)
Hypotonic fluid	Lower sodium than plasma (0.45% saline, 0.18% saline with 4% glucose, half-strength Darrow's, 5% dextrose)
Hyponatraemia	Na < 135 mmol/L (moderate 125–129, severe < 125)
Hypernatraemia	Na > 145 mmol/L (moderate 150–169, severe ≥ 170)
Hypokalaemia	K < 3.5 mmol/L (mild 3.0–3.4, moderate 2.5–2.9, severe 2.0–2.4, critical < 2.0)
Hyperkalaemia	K > 5.5 mmol/L (severe ≥ 7, or any level with ECG changes)
Oliguria	Urine output < 0.5 mL/kg/h (normal ≥ 1 mL/kg/h; ≥ 2 mL/kg/h in infants)

2. Burden and Uganda Context

- Fluid and electrolyte problems are not a disease but they complicate the **commonest killers** of Ugandan children: **diarrhoea** (dehydration, hypo- and hypernatraemia, hypokalaemia), **severe malaria** (acidosis, AKI, hyperkalaemia), **meningitis and pneumonia** (SIADH, hyponatraemia), **SAM** (low potassium and magnesium, refeeding), **AGN, HUS and AKI** (overload, hyperkalaemia), **DKA**, **burns**, and **Burkitt lymphoma** (tumour lysis: hyperkalaemia, hyperphosphataemia).
- The **FEAST trial** (2011), run largely in **Uganda (Mulago, Mbale, Soroti)**, Kenya and Tanzania, showed that **fluid boluses increased mortality** in febrile children with impaired perfusion (mostly malaria and sepsis) who were not hypotensive. It changed WHO practice (ETAT 2016).

UGANDA CONTEXT

On most Ugandan paediatric wards you will find **0.9% saline, Ringer's lactate, 5% and 10% dextrose and 50% dextrose ampoules**, sometimes **half-strength Darrow's in 2.5% dextrose**, and **KCl 15% ampoules** (2 mmol/mL). **Infusion pumps and burettes are scarce**: most children get fluids by gravity through a **20 drops/mL** giving set, which makes overload easy. Electrolytes are available at **Mulago, Mengo, the regional referral hospitals and private labs**, often not at night and rarely at HC IV. **3% saline for IV use** is not routinely stocked outside national/regional referral hospitals. Plan for these limits: use burettes or **micro-drip (60 drops/mL) sets** for small children, write **hourly volumes**, and keep a **fluid chart**.

3. Principles and Framework

Body water

Age	Total body water (% weight)	Extracellular fluid (% weight)
Preterm newborn	~85%	~50–60%
Term newborn	~75–80%	~40–45%
1 year	~60–65%	~25–30%
Adolescent/adult	~60%	~20%

The infant's large ECF is lost first in dehydration, and their turnover is about **3 times** an adult's per kg.

Where Holliday–Segar comes from

In 1957 Holliday and Segar linked water need to **energy use**: about **100 mL of water per 100 kcal** burned. Energy use per kg falls as children grow, so water per kg falls too: **100 mL/kg/day** for the first 10 kg, **50** for the next 10, **20** for each kg above 20. Daily electrolyte needs are roughly **Na 2–3 mmol/kg/day** and **K 1–2 mmol/kg/day**.

Why sick children get hyponatraemic

Pain, nausea, fear, surgery, **pneumonia, meningitis, encephalitis, bronchiolitis** and positive-pressure breathing all release **ADH**, independent of plasma tonicity. The kidney retains free water. If the child is also given **hypotonic fluid**, the extra water dilutes the sodium. That is why NICE NG29 (2015, updated 2020) moved to **isotonic maintenance fluids** and why many sick children need **restricted** volumes.

Sick child: meningitis, pneumonia, pain, surgery

Hypotonic IV fluid, e.g. plain 5% dextrose

↓

Non-osmotic ADH release

Extra free water given

↓

Kidney retains free water

↓

Plasma sodium falls

↓

Water enters brain cells

↓

Cerebral oedema: headache, vomiting, lethargy

↓

Seizures, coma, brainstem herniation

Sodium and the brain

When plasma sodium falls, water enters brain cells (**cerebral oedema**: headache, vomiting, lethargy, seizures, coma). Within 24–48 h the brain adapts by losing intracellular osmoles. After that adaptation, **rapid correction** of chronic hyponatraemia shrinks the cells and causes **osmotic demyelination**. In **hypernatraemia** the brain makes "idiogenic osmoles"; lowering sodium fast then drives water into brain cells (**cerebral oedema and seizures**). **Both errors come from going too fast.**

Potassium

About **98% of potassium is inside cells**; serum K is a poor guide to total stores. Insulin, β-agonists and alkalosis drive K **into** cells; acidosis, cell breakdown (haemolysis, tumour lysis, rhabdomyolysis) and insulin deficiency move it **out**. The heart is the target organ: **low K** → flat T waves, U waves, ectopics; **high K** → peaked T waves, wide QRS, sine wave, VF/asystole.

The four reasons to give fluid

Purpose	Question	Fluid and volume
Resuscitation	Is the child in shock (all three signs)?	Isotonic bolus (section 4, step 4); special rules in SAM, severe anaemia, heart failure
Deficit	How dehydrated?	% × kg × 10 mL, over 24 h (48 h or more in hypernatraemia); WHO plans for diarrhoea
Maintenance	Can the child drink or feed?	Holliday–Segar, isotonic with glucose, often restricted to 2/3
Ongoing losses	What is still being lost?	Measure and replace volume for volume

Composition of common fluids (mmol/L)

Fluid	Na	K	Cl	Other	Glucose	Use
0.9% saline ("normal saline")	154	0	154		0	Bolus, deficit, maintenance base
Ringer's lactate / Hartmann's	130–131	4–5	109–111	Lactate 28–29, Ca 2	0	**First choice for resuscitation and Plan C**

Fluid	Na	K	Cl	Other	Glucose	Use
Plasma-Lyte 148	140	5	98	Mg 1.5, acetate, gluconate	0	Bolus/maintenance (private/NRH)
5% dextrose	0	0	0		50 g/L	**Not for maintenance alone**; vehicle for drugs
10% dextrose	0	0	0		100 g/L	Hypoglycaemia (5 mL/kg), neonates
0.45% saline + 5% dextrose	77	0	77		50 g/L	Hypotonic; slow phase of hypernatraemia; SAM shock
Half-strength Darrow's + 2.5% dextrose	61	17	52	Lactate 27	25 g/L	Old UCG option; hypotonic
Low-osmolarity ORS	75	20	65	Citrate 10	75 mmol/L	Oral rehydration
ReSoMal	45	40		Mg, Zn, Cu		SAM rehydration
15% KCl ampoule		**2 mmol/mL**				**Additive only; never undiluted**

Normal vital signs (WHO / ETAT)

Age	RR (fast if ≥)	HR (awake, normal)	Systolic BP (lower limit approx.)
< 2 months	60	110–160	60
2–12 months	50	100–160	70
1–5 years	40	90–140	70 + (2 × age in years)
5–12 years	30	70–120	70 + (2 × age), up to 90

4. Application in Practice

Step 1: First 5 minutes

1. **ETAT**: airway, breathing (SpO_2, give oxygen if < 90%), **circulation** (cold hands, CRT, pulse), coma, convulsions, **severe dehydration**.
2. **Weigh** (or estimate: (age in years + 4) × 2 for 1–10 years, as a last resort).
3. **IV or IO access**; draw **glucose, Hb/HemoCue, electrolytes and creatinine** (if the lab can run them), malaria RDT, group and cross-match if pale.
4. **Glucose < 2.5 mmol/L** (WHO; **< 3 mmol/L in SAM**; UCG 2023 uses < 3.0) → **10% glucose 5 mL/kg IV**.
5. **Shock** → bolus (step 4), unless SAM (15 mL/kg over 1 h) or severe anaemia (blood) or heart failure (no bolus).
6. **Convulsing** → glucose, benzodiazepine, then think sodium.
7. **Stop** any running hypotonic fluid and any potassium infusion while you assess.

Step 2: Focused history

Each question has a reason.

Ask	Why it matters
Weight (today and recent, from the child health card)	Every calculation depends on it; weight change = water change; % dehydration
Intake: breastfeeding, drinking, feeds; what fluids at home; formula preparation	Under-intake, dilute feeds (hyponatraemia) or concentrated formula (hypernatraemia)
Losses: stools, vomiting (bilious?), urine output, sweating, burns, drains	Size and type of loss decide deficit and replacement fluid type
Urine: last void, number of wet nappies, colour, polyuria	Oliguria = dehydration or AKI, and whether potassium is safe; polyuria with dehydration = DKA, diabetes insipidus, salt-wasting
Fluids already given (type, volume, time)	Avoid double dosing and overload; total fluid so far
Neurology: irritability, lethargy, headache, convulsions	Sodium disorders, hypoglycaemia, meningitis; urgency of sodium correction
Weakness, abdominal distension, palpitations	Hypokalaemia (and hyperkalaemia)
Underlying illness: malaria, meningitis, pneumonia, kidney disease (puffy face, cola urine), heart disease, SAM, diabetes, Burkitt or other cancer on chemotherapy	Predicts the disorder (e.g. AGN → overload, hyperkalaemia); restriction and special protocols
Medicines: diuretics, salbutamol, ACE inhibitors, amphotericin, chemotherapy, IV fluids already given	Iatrogenic causes; K and Na effects
Nutrition, HIV status, immunisation	SAM changes all fluid rules; HIV and missed vaccines change the differential

Step 3: Focused examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: introduce yourself, gain consent from the parent and assent from the child, **hand hygiene**, position the child on the parent's lap or the bed and expose as needed; observe before touching.

1. **General inspection (end of the bed)**: well or unwell, conscious level (AVPU or Blantyre coma scale), breathing pattern (deep acidotic breathing; rising RR in overload), puffiness of the face, visible wasting or oedema, dysmorphism (ambiguous genitalia: CAH).
2. **Vital signs and growth**: **weight** (same scale), temperature, **HR, RR, BP, SpO_2, CRT** against the age table in section 3 (hypotension is late in children); MUAC, WLZ, oedema of both feet.
3. **Hands**: temperature, **capillary refill** (> 3 s abnormal), radial pulse volume and rate, pallor.
4. **Face**: eyes (sunken; **puffy eyelids** in overload), fontanelle (sunken or bulging), mucosae, tears.
5. **Neck**: **raised JVP** (older child), neck stiffness.
6. **Systems**:
   - **Hydration**: skin pinch (doughy in hypernatraemia).
   - **Cardiovascular**: **gallop**, murmurs.
   - **Respiratory**: basal **crackles**.
   - **Abdomen**: **liver span and edge** (a **tender enlarging liver** is the best overload sign in infants), distension and absent bowel sounds (ileus, hypokalaemia), masses (Burkitt), bladder.
   - **Neurological**: tone (floppy in hypokalaemia), reflexes.
7. **Complete**: thank the parent and child, hand hygiene, and summarise the volume state (dry, euvolaemic or overloaded) and the fluid plan.

Document these positives	And these negatives
Weight and time weighed	"No gallop, no crackles, liver 1 cm"

Document these positives	And these negatives
CRT, pulse volume, BP	"No oedema; MUAC 13.5 cm"
Hydration signs	"Passed urine at 14:00"
Conscious level (BCS/AVPU)	"Fontanelle not bulging; no neck stiffness"

Step 4: Resuscitation (is the child in shock?)

Shock = cold hands + CRT > 3 s + weak, fast pulse (all three; WHO ETAT 2016).

Situation	Fluid	Volume and rate	Then
Shock, not malnourished, not anaemic	**Ringer's lactate or 0.9% saline** (never glucose-only)	**UCG 2023: 20 mL/kg as rapidly as possible**; WHO ETAT 2016: **10–20 mL/kg over 30–60 min**; NICE NG29: 10 mL/kg over < 10 min	Reassess after each bolus; repeat 10–20 mL/kg if still shocked; involve a senior after 20 mL/kg; consider inotropes (adrenaline/dopamine) and septic shock after 40–60 mL/kg
Impaired circulation (1–2 signs only), febrile illness, severe malaria, pneumonia, meningitis	**No bolus** (FEAST)	**Maintenance** fluids	Reassess within 1 hour
Severe dehydration from diarrhoea	Ringer's lactate	**WHO Plan C**: 30 mL/kg then 70 mL/kg (see Diarrhoea topic)	Reassess, ORS
SAM with shock	RL with 5% glucose, half-strength Darrow's with 5% glucose, or 0.45% saline with 5% glucose	**15 mL/kg over 1 hour**, repeat once if improving	Oral/NG ReSoMal; if no improvement, **blood 10 mL/kg over 3 h**
Shock with severe anaemia (Hb < 5 g/dL)	**Whole blood 20 mL/kg** (or packed cells 10 mL/kg)	As early as possible	Other IV fluids only to maintain hydration
Haemorrhage	Blood (UCG: 20 mL/kg) after initial crystalloid		Stop the bleeding
DKA with shock	0.9% saline	ISPAD 2022: **10–20 mL/kg over 20–30 min** (repeat if needed)	Deficit over 24–48 h, insulin after the first hour
Cardiogenic shock / heart failure	**No bolus**; small cautious 5–10 mL/kg only on senior advice		Oxygen, furosemide, inotropes

RED FLAG

Stop the bolus if you hear **crackles**, a **gallop**, the **liver edge moves down**, the eyelids swell, or the child becomes more breathless or less conscious. **Never** give glucose-only fluids as a bolus (hyperglycaemia and hyponatraemia).

Step 5: Maintenance fluids

Volume (Holliday–Segar, NICE NG29, WHO Pocket Book):

Weight	Per day	Per hour ("4-2-1")
First 10 kg	**100 mL/kg**	**4 mL/kg/h**

Weight	Per day	Per hour ("4-2-1")
Next 10 kg (11–20 kg)	**+ 50 mL/kg**	**+ 2 mL/kg/h**
Each kg above 20 kg	**+ 20 mL/kg** (UCG: 25 mL/kg)	**+ 1 mL/kg/h**
Cap	About **2–2.5 L/day** (adult maintenance)	

Adjustments:

- **Fever**: + about **10% per °C** above 37 °C (UCG).
- **Restrict to 2/3 (NICE: 50–80%)** in conditions with high ADH: **meningitis**, **pneumonia**, **bronchiolitis**, **head injury**, **post-operative**, encephalitis, ventilated children.
- **AGN / AKI / oliguria**: insensible losses (**about 400 mL/m²/day**, roughly **15–25 mL/kg/day** in small children) **plus the previous day's urine output**.
- **Heart failure**: 50–70% of maintenance with furosemide.
- **Subtract** oral, NG and drug volumes from the IV total.
- Change to the **oral route** as soon as the child can drink or take NG feeds.

Neonates (UCG 2023):

Day of life	Total fluid (oral and/or IV)
Day 1	**60 mL/kg/day** of 10% glucose
Day 2	**90 mL/kg/day** of 10% glucose
Day 3	**120 mL/kg/day** (half-normal saline with 5% glucose, or 10% glucose with added sodium per local protocol)
Day 4 onwards	**150 mL/kg/day**; up to **180 mL/kg/day** once oral feeding is established

Do not exceed 100 mL/kg/day as IV fluid alone unless the baby is dehydrated or under phototherapy or a radiant heater (UCG). Breastfeed or give expressed breast milk by NG tube whenever possible.

Type of maintenance fluid:

Child	Fluid
1 month and older (NICE NG29, RCH)	**Isotonic with glucose**: **0.9% saline with 5% glucose** (± KCl 20 mmol/L) or **Ringer's lactate with 5% glucose**; Plasma-Lyte 148 with 5% glucose where available
Term neonate ≥ 8 days	Isotonic crystalloid with **5–10% glucose**
Neonate first days	**10% glucose** (sodium added from day 2–3 when urine output is established)
SAM	**Avoid IV maintenance**; feed F-75 orally/NG
Older UCG practice	One-third 0.9% saline and two-thirds 5% dextrose (adult chapter) or half-strength Darrow's in 2.5% glucose: **hypotonic**; check sodium daily if used

UGANDA CONTEXT

Making "0.9% saline with 5% glucose" on a Ugandan ward: remove **50 mL** from a **500 mL bag of 0.9% saline** and add **50 mL of 50% dextrose** → about **5% glucose in 0.9% saline**. **10% glucose** from stock: **1 part 50% + 4 parts water**, or **1 part 50% + 9 parts 5% dextrose** (UCG). Label the bag clearly with what you added, the time and your name.

Step 6: Deficit and ongoing losses

1. **Deficit (mL) = % dehydration × weight (kg) × 10.** (5% of 10 kg = 500 mL.)

2. Replace the deficit **over 24 h** in most children (RCH: no more than about **5% deficit replaced IV in 24 h**, then reassess); **over 48–72 h** in hypernatraemia; diarrhoeal dehydration uses the **WHO plans** instead.
3. **Ongoing losses**: measure and replace **volume for volume** every 4–6 h: diarrhoea with **ORS** or Ringer's lactate; **NG/vomit losses with 0.9% saline + 10–20 mmol/L KCl**; burns per a burns formula (e.g. Parkland 3–4 mL/kg/% burn in 24 h plus maintenance).

Step 7: Bedside and laboratory tests

Bedside

Test	How	What it tells you
Weight	Same scale daily (or 12-hourly)	Fluid gain/loss: 1 kg ≈ 1 L
RBS	Glucometer	Hypo- or hyperglycaemia (DKA); glucose in fluids needed?
Urine output	Weigh nappies (1 g = 1 mL) or bag	Normal ≥ **1 mL/kg/h** (≥ 2 mL/kg/h in infants); < 0.5 mL/kg/h = oliguria
Urine dipstick and SG	Dipstick	SG > 1.020 concentrated (dehydration, SIADH); glucose and ketones (DKA); blood and protein (AGN)
HemoCue Hb	Finger prick	Severe anaemia changes the shock plan
ECG / cardiac monitor	3-lead monitor or 12-lead	Peaked T waves, wide QRS (high K); flat T, U waves (low K)
Point-of-care electrolytes / blood gas	i-STAT or ward gas (if available)	Na, K, glucose, lactate, pH, bicarbonate in minutes
Fluid balance chart	Hourly input and output	Running total guides the next prescription

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Serum Na, K, Cl, bicarbonate	Before IV fluids (except short elective cases), then at least **every 24 h**; **4–6 h** after starting if very sick or abnormal	Na < 135 or > 145; K < 3.5 or > 5.5; low bicarbonate	Choice of fluid, rate of correction, potassium dose	**RRH/NRH, Mengo, private labs**; rarely at HC IV
Urea and creatinine	Dehydration, oliguria, AGN, malaria with dark urine, HUS	Raised (pre-renal or intrinsic AKI)	Fluid restriction, dialysis referral (peritoneal dialysis at Mulago)	RRH/NRH
Serum and urine osmolality, urine sodium	Unexplained hyponatraemia	Urine Na > 20–30 mmol/L with concentrated urine = SIADH or renal salt loss; < 20 = extrarenal loss	Restrict (SIADH) vs replace salt (salt-wasting)	NRH, private labs
Blood gas / lactate	Shock, DKA, severe malaria, sepsis	pH < 7.3, bicarbonate < 15, lactate > 4	Severity; response to treatment	NRH/ICU
Magnesium, calcium, phosphate	Refractory hypokalaemia, SAM refeeding, tumour lysis, tetany	Low Mg; low Ca; low PO_4 (refeeding) or high PO_4 (tumour lysis)	Replace Mg before K corrects; slow feeds	RRH/NRH
Glucose (lab)	DKA, hypoglycaemia	> 11 or < 3 mmol/L	Insulin protocol or glucose	HC IV and above

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	Pallor, sepsis	Severe anaemia; haemoconcentration	Transfuse; sepsis care	HC IV and above
Chest X-ray	Suspected overload or heart failure	Cardiomegaly, pulmonary oedema	Stop fluids, furosemide	General Hospital and above

Step 8: Monitoring every child on IV fluids

1. **Prescribe in mL/h** with the fluid type, additives and duration. Review the prescription **at least every 12–24 h**.
2. **Fluid balance chart**: all input (IV, oral, NG, drugs) and output (urine, stool, vomit, drains) every hour; running totals.
3. **Weigh daily** (12-hourly in infants, SAM, AGN).
4. **Electrolytes and glucose** before starting (except most elective surgery) and **at least every 24 h**; every **4–6 h** if sick, abnormal or correcting sodium.
5. **Clinical review** at least every 4 h: hydration, overload signs, conscious level, urine output.
6. **Switch to oral** as soon as possible, and take the line out.

Putting it together

Model one-line summary

> "A 3-year-old boy, 13 kg, day 2 of treatment for bacterial meningitis, on 'D5 1 L 8-hourly' since admission, who had a generalised seizure this morning; Na 121 mmol/L, glucose 6.4 mmol/L: acute symptomatic hyponatraemia from hypotonic fluid plus SIADH, with fluid overload (3 L/day against a maintenance of 1150 mL)."

Problem list

1. Acute symptomatic hyponatraemia (Na 121) with seizure.
2. Iatrogenic fluid overload with a hypotonic fluid.
3. Bacterial meningitis (SIADH risk) on ceftriaxone.

Working diagnosis and plan: stop D5; **2 mL/kg 3% saline (26 mL) over 10–15 min**, repeat once if still seizing; then **isotonic fluid with glucose at 2/3 maintenance**; Na **2–4 hourly**; aim for a rise of **no more than 8 mmol/L in 24 h** after symptoms stop.

Model case presentation (ward round)

> "Sarah is a **3-year-old girl, 13 kg** (WAZ −0.8, WHZ −0.5, MUAC 15 cm, no oedema), **fully immunised** on the UNEPI schedule, **HIV-negative**, eating family foods, admitted 2 days ago with **bacterial meningitis** on ceftriaxone. Since admission she has been on **5% dextrose, 500 mL 6-hourly (2 L/day)**. This morning she had a **generalised tonic-clonic seizure** for 4 minutes, stopped with rectal diazepam. Now **BCS 4/5**, HR 128, RR 32, BP 98/60, CRT 2 s, **puffy eyelids**, liver 2 cm, no crackles. RBS 6.4. **Na 121, K 3.4**, creatinine normal. **Summary**: acute symptomatic hyponatraemia from a **hypotonic fluid at nearly twice maintenance** on top of **SIADH from meningitis**. Plan: **3% saline 26 mL over 15 min**, stop D5, **0.9% saline with 5% glucose + KCl 20 mmol/L at 2/3 maintenance (35 mL/h)**, Na 4-hourly, aiming for **≤ 8 mmol/L rise in 24 h**, daily weight, fluid chart, and an incident report."

EXAM FAVOURITE

Worked example 1: Holliday–Segar.

- **7 kg** infant: 7 × 100 = **700 mL/day** = 29 mL/h (or 7 × 4 = 28 mL/h).
- **15 kg** child: (10 × 100) + (5 × 50) = 1000 + 250 = **1250 mL/day** = 52 mL/h (or 40 + 10 = 50 mL/h).
- **32 kg** child: 1000 + 500 + (12 × 20) = **1740 mL/day** = 72 mL/h (or 40 + 20 + 12 = 72 mL/h). **UCG 2023 variant** (25 mL/kg above 20 kg): 32 kg = 1000 + 500 + 300 = 1800 mL/day. **2/3 maintenance** for the 15 kg child with meningitis = 1250 × 2/3 ≈ **830 mL/day ≈ 35 mL/h**.

EXAM FAVOURITE

Worked example 2: deficit plus maintenance, and drops per minute. A 12 kg child is 7% dehydrated, cannot drink and needs IV fluids (not shocked).

- Deficit = 7 × 12 × 10 = **840 mL**; maintenance = 1000 + 100 = **1100 mL**; total **1940 mL over 24 h ≈ 81 mL/h**.
- With a **20 drops/mL** giving set: drops/min = mL/h × 20 ÷ 60 = 81 ÷ 3 ≈ **27 drops/min**.
- With a **60 drops/mL** micro-drip/burette: drops/min = mL/h = **81 drops/min**.
- Add **KCl 20 mmol/L** once he passes urine (15% KCl = 2 mmol/mL → **10 mL into 1 L**, or 5 mL into a 500 mL bag; mix well).
- Weigh again in 12 h; recheck Na and K at 24 h. (For diarrhoeal dehydration use the WHO plans instead; see the Diarrhoea topic.)

5. Clinical Acumen

Red flags

RED FLAG

Shock; **convulsions or reduced consciousness** in a child on IV fluids (check glucose and sodium **now**); **Na < 125 or > 160 mmol/L**; **K > 6.5 or < 2.5 mmol/L**, or any K abnormality with **ECG changes**; **no urine for 6–8 h** on fluids; **signs of overload** (gallop, crackles, enlarging liver, puffy eyes); **SAM, severe anaemia, heart disease or AGN** (fluid will kill); and **any child on IV fluids for more than 24 h without an electrolyte check**.

Thinking like a clinician

- **First ask "is this child in shock?"** Only shock earns a bolus. Fast heart rate from fever, crying or malaria, or cool hands alone, is not shock.
- **If shock, ask "why?"** Hypovolaemia (diarrhoea, bleeding, burns), sepsis, **severe anaemia** (blood, not saline), **SAM** (15 mL/kg slowly), **cardiogenic** (bolus worsens it: big liver, gallop, crackles), DKA (cautious bolus).
- **Calculate, then round down for a sick child.** Meningitis, pneumonia, bronchiolitis, post-op and head injury: **2/3 maintenance** (NICE approach; in meningitis without hyponatraemia or raised ICP, WHO and Cochrane 2016 favour full isotonic maintenance: see Meningitis). AGN and AKI: **insensible losses (about 400 mL/m²/day, roughly 15–25 mL/kg/day) plus urine output**.
- **The best monitor is the scale.** A child who gains more than 5% on IV fluids is overloaded.
- **Seizure on IV fluids = glucose, then sodium.** Hypoglycaemia is quicker to fix; hyponatraemia is the commonest hidden cause.
- **Potassium: "no pee, no K"**. Never add KCl until urine is flowing, never inject KCl as a push, and **mix the bag well** after adding it.

- **Interpret electrolytes in order**: comment on the **volume state first**, then sodium, potassium, bicarbonate, urea/creatinine and glucose, and link each to the story.
- **Common student mistakes**: using plain D5 for maintenance; bolusing a febrile child with malaria who is not shocked; forgetting to subtract oral/NG feeds from the IV volume; writing "1 L 8-hourly" for a 6 kg infant (that is 3 L/day, five times maintenance); correcting chronic hypernatraemia in a few hours.

Causes of each disorder in Ugandan children

Disorder	Common causes	Distinguishing features	Key investigation
Hypovolaemia	Diarrhoea and vomiting, cholera, burns, haemorrhage, DKA, sepsis (capillary leak)	Dehydration signs, low urine output, weight loss	Weight, urea, Na
Fluid overload	Excess IV fluid, **AGN**, AKI, heart failure (rheumatic, congenital), severe anaemia, SAM refeeding	Puffy eyes, oedema, big liver, gallop, crackles, weight gain	CXR, creatinine, urine
Hyponatraemia, hypovolaemic	Diarrhoea replaced with water or dilute fluids, salt-wasting CAH, diuretics, cerebral salt wasting	Dehydrated; high K suggests **CAH or adrenal crisis**	Urine Na, K, glucose, 17-OHP (CAH)
Hyponatraemia, euvolaemic	**SIADH** (meningitis, pneumonia, TB meningitis, encephalitis, post-op), **hypotonic IV fluids**, water intoxication (diluted formula)	Normal hydration, concentrated urine, urine Na high	Urine and serum osmolality, urine Na
Hyponatraemia, hypervolaemic	**Nephrotic syndrome**, heart failure, cirrhosis, AKI	Oedema	Urine protein, albumin, echo
Hypernatraemia	Diarrhoea with high fever, concentrated formula, poor breastfeeding in a newborn, salt poisoning, diabetes insipidus (after meningitis or head injury), excess sodium bicarbonate	Very irritable, doughy skin, thirst; polyuria in DI	Urine osmolality, Na
Hypokalaemia	**Diarrhoea**, vomiting (pyloric stenosis: low Cl, alkalosis), **SAM**, refeeding, **DKA treatment**, salbutamol, diuretics, amphotericin, hypomagnesaemia	Floppy, ileus, ECG changes	K, Mg, bicarbonate
Hyperkalaemia	**AKI** (AGN, HUS, malaria, dehydration), **haemolysed sample** (heel prick), **tumour lysis (Burkitt)**, haemolysis, **CAH**, rhabdomyolysis, massive transfusion, K in fluids, ACE inhibitors	Oliguria, ECG changes	Repeat free-flowing sample, creatinine, ECG

Scenarios

- **A child on IV fluids is drowsy.** Glucose (bedside), then sodium; look at the fluid chart (type and volume); look for meningitis or raised ICP.
- **Is this child overloaded?** Weight gain, puffy eyelids, oedema, enlarging tender liver, gallop, crackles, rising RR and falling SpO_2, raised JVP in older children.
- **K 6.8 from a heel prick in a well baby with a normal ECG.** Repeat a free-flowing sample before treating.

6. Common Problems and Their Management

Hyponatraemia (Na < 135 mmol/L)

Step 1: Is the child symptomatic? (Headache, vomiting, lethargy, confusion, **seizures**, reduced consciousness.)

Acute symptomatic hyponatraemia (emergency):

1. ABC; oxygen; treat seizure (benzodiazepine) and check **glucose**.
2. **Hypertonic saline: 2 mL/kg of 2.7% (or 3%) NaCl IV over 10–15 min (max 100 mL)** (NICE NG29). Repeat once if symptoms persist; check Na and consider a **third dose** if still symptomatic. Each 1 mL/kg of 3% raises Na by about **1 mmol/L**; a rise of **4–6 mmol/L** usually stops seizures.
3. **Stop hypotonic fluids**; continue **isotonic** fluid at a **restricted** rate.
4. Once symptoms stop: total rise **≤ 8 mmol/L in 24 h** (Medscape, RCH 6–8; **NICE: never more than 12 mmol/L in 24 h**). Check Na every **2–4 hours**.

Asymptomatic or mild:

Cause (volume state)	Treatment
Hypovolaemic (diarrhoea, salt loss)	Rehydrate with **isotonic fluid** (ORS, RL or 0.9% saline); Na rises as volume is restored. Watch for an overshoot once ADH switches off (brisk urine)
Euvolaemic / SIADH (meningitis, pneumonia, post-op, hypotonic fluids)	**Restrict** to 50–2/3 maintenance; **isotonic** fluid only; treat the cause. **Do not manage acute hyponatraemic encephalopathy by fluid restriction alone** (NICE)
Hypervolaemic (nephrotic, heart failure, AKI)	Restrict fluid and salt; diuretics (furosemide 1 mg/kg) for overload; treat the cause
Salt-wasting CAH / adrenal crisis (hyponatraemia + **hyperkalaemia** + hypoglycaemia in an infant; ambiguous genitalia in girls)	0.9% saline 20 mL/kg if shocked, **hydrocortisone**, glucose; specialist referral

EXAM FAVOURITE

Worked example 3: hyponatraemia. A 10 kg child with pneumonia, on "D5 500 mL 6-hourly" (2 L/day; maintenance 1 L), seizes. Na **118 mmol/L**, glucose 7.

- Give **3% saline 20 mL (2 mL/kg) over 10–15 min**; seizure stops; repeat Na 30 min later = **122**.
- Change to **0.9% saline with 5% glucose at 2/3 maintenance = 660 mL/day ≈ 28 mL/h**.
- Target Na at 24 h: no higher than **118 + 8 = 126 mmol/L**. If it rises faster (brisk urine as ADH falls), slow the saline and give some 5% glucose on senior advice.

Hypernatraemia (Na > 145 mmol/L)

1. **Treat shock first** with **0.9% saline or RL boluses** (isotonic fluid is relatively hypotonic to this child's plasma).
2. Then correct **slowly**: sodium should fall **no faster than 0.5 mmol/L per hour** (about **10–12 mmol/L in 24 h**; NICE: never more than 12 mmol/L in 24 h). Replace the deficit over **48 hours or more** (72 h if Na > 160–170).
3. **Fluid**: **ORS by mouth or NG** where possible (NICE: ORS is also suitable in hypernatraemia). IV: start with **0.9% saline (with glucose)**; if sodium does not fall, move to **0.45% saline with 5% glucose** (NICE algorithm).
4. **Check Na every 2–4 hours** at first (RCH: 1–2 h after starting, then 4–6-hourly). **If Na falls > 0.5 mmol/L/h, slow or stop the fluid.**
5. Add **KCl 20–40 mmol/L** once the child passes urine (helps water enter cells; Medscape).
6. **Seizures during correction** = cerebral oedema: stop hypotonic fluid, give **3% saline 2–3 mL/kg** (or mannitol) and call ICU.
7. **Fix the cause**: correct formula preparation (1 level scoop per 30 mL water), support breastfeeding in a newborn (weigh, observe a feed), treat fever, test urine for **diabetes insipidus** (dilute urine despite hypernatraemia).

EXAM FAVOURITE

Worked example 4: free-water deficit (rough guide). Free water deficit (L) = 0.6 × weight × (current Na ÷ 145 − 1). A 6 kg infant with Na 166: 0.6 × 6 × (166/145 − 1) = 3.6 × 0.145 ≈ **0.52 L**. Spread over **48 h** with maintenance (600 mL/day × 2 = 1.2 L) → about **1.7 L in 48 h ≈ 36 mL/h**, then **measure Na** and adjust. Formulas only start you off; **serial sodium values drive the rate**.

Hypokalaemia (K < 3.5 mmol/L)

1. **Confirm** (free-flowing venous sample), check **Mg**, bicarbonate, glucose; **ECG** if K < 3.0 or symptoms.
2. **Ensure the child is passing urine.**
3. **Oral route preferred** (RCH): **KCl 1–2 mmol/kg/dose (max 20 mmol/dose)**; repeat after checking K, to a **maximum 5 mmol/kg/day (max 50 mmol/day)**. Maintenance if needed **2–5 mmol/kg/day** in divided doses. Slow-K tablet = 8 mmol; give with food.
4. **IV** if unable to take oral, K < 2.5 or ECG changes:
   - **Add KCl to maintenance fluid: 20 mmol/L** (10 mL of 15% KCl per 1 L), up to **40 mmol/L** in a peripheral line.
   - **Acute replacement on a general ward: 0.2 mmol/kg/h for 3 hours (max 10 mmol/h)**, ECG monitoring if K < 3; repeat K 1 hour after.
   - **Higher rates (0.4 mmol/kg/h, max 20 mmol/h)** only in **ICU** via a central line with continuous monitoring.
   - **Never** give KCl as an IV push or IM. **Invert the bag several times** after adding KCl.
5. **Correct magnesium** if low (hypokalaemia is refractory until Mg is replaced).
6. **SAM**: extra potassium **3–4 mmol/kg/day** and magnesium **0.4–0.6 mmol/kg/day**, given in feeds or ReSoMal (the combined electrolyte/mineral mix), not IV.

Hyperkalaemia (K > 5.5 mmol/L)

1. **Is it real?** Repeat a **free-flowing venous sample** (heel pricks and squeezed fingers haemolyse); but **do not delay treatment** if ECG changes or K ≥ 7.
2. **Stop all potassium** (IV additives, oral supplements, RL is low in K but stop it in severe AKI), stop ACE inhibitors and K-sparing diuretics. **Cardiac monitor and ECG.**
3. **Treat in order:**

Step	Drug	Dose	Route / rate	Onset / duration	Notes
1. Protect the heart (ECG changes or K ≥ 7)	**Calcium gluconate 10%**	**0.5 mL/kg (max 20 mL)**; RCH 0.68 mL/kg (0.15 mmol/kg), max 30 mL	IV/IO over **2–5 min** if unstable, 15–20 min if stable	< 3 min; lasts ~30 min	Does **not** lower K; repeat if ECG still abnormal; not in the same line as bicarbonate; avoid if on digoxin
2. Shift K into cells	**Salbutamol**	**2.5 mg (≤ 25 kg); 5 mg (> 25 kg)**	Nebulised, repeat 1–2-hourly	30 min; 2–3 h	Quick, available everywhere in Uganda
	Insulin (soluble/Actrapid) + glucose	**Insulin 0.1 unit/kg IV (max 10 units) + 10% glucose 5 mL/kg IV**	Bolus, then infusion if needed	15 min; 2–3 h	**Check glucose every 30–60 min** (hypoglycaemia risk)
	Sodium bicarbonate 8.4%	**1 mmol/kg (1 mL/kg)**	Slow IV over 30 min (dilute 1:1 in neonates and infants)	30–60 min	**Only if metabolic acidosis**; not with calcium

Step	Drug	Dose	Route / rate	Onset / duration	Notes
3. Remove K from the body	**Calcium/sodium polystyrene sulfonate (Resonium)**	**0.5–1 g/kg (max 30 g)**	Oral or rectal, 6-hourly	Hours	Mild effect; constipation
	Furosemide	**1 mg/kg** (max 40 mg)	IV		Only if kidneys produce urine
	Dialysis		Peritoneal or haemodialysis		Refractory hyperkalaemia, AKI: **Mulago (PD), some RRHs**

4. **Treat the cause**: AKI (AGN, HUS, malaria), tumour lysis (hydration, allopurinol, avoid K), adrenal crisis (hydrocortisone), haemolysis.

> NOTE
> The highYield list gives calcium gluconate "over 5–10 min"; the table above gives **2–5 min if unstable, 15–20 min if stable** (source notes). Both are within published ranges Verify *rate against the current RCH hyperkalaemia guideline*.

> DRUG DOSING
> **Potassium safety rules**: **15% KCl = 2 mmol/mL**; **10 mL into 1 L = 20 mmol/L**; **maximum 40 mmol/L peripherally**; **maximum 0.2 mmol/kg/h on a general ward (max 10 mmol/h)**; **never undiluted, never IV push, never IM**; only once urine output is established. Store KCl ampoules separately from saline and water ampoules: look-alike ampoules have killed children.

Fluid overload

Recognise it (weight gain > 5%, puffy eyelids, oedema, enlarging tender liver, gallop, crackles, rising RR, falling SpO_2, raised JVP); **stop or reduce IV fluids**, sit the child up, oxygen, **furosemide 1 mg/kg IV**, chest X-ray if unsure; in AGN/AKI restrict to insensible losses plus urine output.

Complications of fluid prescribing errors

Error	Consequence
Too much fluid	Pulmonary oedema, heart failure, raised ICP, death (FEAST), worse outcomes in SAM, AGN, severe anaemia
Hypotonic fluid	**Hyponatraemic encephalopathy**: seizures, brainstem herniation, death
Correcting hyponatraemia too fast	**Osmotic demyelination** (weeks later: quadriparesis, mutism)
Correcting hypernatraemia too fast	**Cerebral oedema**, seizures, death
Too little fluid	Persistent shock, AKI, cerebral venous thrombosis (hypernatraemia), death
Potassium errors	Cardiac arrest (KCl push or overdose), arrhythmias, ileus, paralysis
Glucose errors	Hypoglycaemia (no glucose in maintenance; insulin), hyperglycaemia (glucose boluses)
IV line	Thrombophlebitis, extravasation injury (KCl, calcium, dextrose 10%+), line sepsis

7. Guidelines, Policy and Law in Uganda

Which guideline says what

Topic	UCG 2023	WHO (ETAT 2016, Pocket Book 2013)	NICE NG29 / RCH
Shock bolus	**20 mL/kg RL/NS as rapidly as possible**	**10–20 mL/kg over 30–60 min**; no bolus with only 1–2 signs	10 mL/kg over < 10 min
Maintenance above 20 kg	**25 mL/kg/day**	20 mL/kg/day (Holliday–Segar)	20 mL/kg/day
Maintenance fluid type	Older practice: one-third 0.9% saline + two-thirds 5% dextrose, or half-strength Darrow's in 2.5% glucose (hypotonic)		**Isotonic with glucose**
Hypoglycaemia threshold	< 3.0 mmol/L	< 2.5 mmol/L (< 3 in SAM)	
Neonatal fluids	Day 1 60 → day 4 150 mL/kg/day		

UGANDA CONTEXT

Divergence to flag: UCG 2023 still lists **hypotonic** maintenance options and a faster shock bolus than WHO ETAT 2016. On the ward, use **isotonic fluid with glucose** and check sodium daily if a hypotonic fluid is used; give boluses **only to children with all three signs of shock** (FEAST, WHO ETAT 2016). In meningitis, NICE restricts to 2/3, whereas WHO and Cochrane 2016 favour full isotonic maintenance unless Na is low or ICP is raised. In exams, quote UCG and name the WHO and NICE differences.

What each level can realistically do

Setting	What you can realistically do	Refer up when
HC III	Weigh; ETAT; IV/IO access; RL/NS boluses; WHO Plans; 10% glucose; NG ORS	Any suspected electrolyte disorder, seizures on fluids, oliguria → HC IV or hospital
HC IV / General Hospital	Inpatient IV maintenance with glucose; KCl additive; HemoCue; transfusion where a blood fridge exists; **no electrolyte lab in most**: treat empirically by protocol	Seizures, AKI, Na or K disorder needing measurement → RRH
RRH / Mengo / Mulago (NRH)	Electrolytes, creatinine, ECG, insulin and calcium for hyperkalaemia, 3% saline (NRH/RRH), peritoneal dialysis (Mulago), HDU/ICU	Dialysis or ICU not available at the RRH → NRH

Safety and governance

- **Prescribe fluids like a drug**: weight, type, additives, mL/h, drops/min for the giving set, duration and monitoring, signed and timed.
- **KCl storage**: keep 15% KCl ampoules separate from saline and water ampoules.
- **Label every bag** with additives, time and name.
- **Incident reporting**: a seizure from a hypotonic fluid or a KCl error is a reportable patient-safety incident (as in the model case) Verify *name of the MoH or hospital incident-reporting system*.

8. Key Points

High-yield points

- **Weigh first**; prescribe in mL/h.
- **Only shock (all three signs) earns a bolus**; SAM gets 15 mL/kg over 1 h; severe anaemia gets blood; heart failure gets none.
- **Isotonic with glucose** for maintenance; **never plain 5% dextrose**.
- **2/3 maintenance** for meningitis, pneumonia, bronchiolitis, head injury and post-op.
- **Seizure on fluids: glucose, then sodium.**
- **Sodium: slow in, slow out.**
- **No pee, no K**; max 40 mmol/L peripherally and 0.2 mmol/kg/h on the ward.
- **Calcium protects the heart but does not lower K.**
- **The scale is the best monitor**; > 5% gain = overload.
- **Electrolytes before and at least every 24 h** on IV fluids.

Numbers & doses

Item	Value
Maintenance (Holliday–Segar)	**100 / 50 / 20 mL/kg/day** (UCG: 25 above 20 kg); **4 / 2 / 1 mL/kg/h**
Fever	+ **10% per °C**
Restricted maintenance (high ADH)	**2/3** (NICE **50–80%**)
AGN/AKI	Insensible **400 mL/m²/day** (≈ 15–25 mL/kg/day) + urine output
Neonatal fluids (UCG)	Day 1 **60**, day 2 **90**, day 3 **120**, day 4+ **150 mL/kg/day** (oral up to 180)
Shock bolus	UCG **20 mL/kg RL/NS rapidly**; WHO ETAT **10–20 mL/kg over 30–60 min**; NICE **10 mL/kg < 10 min**
SAM shock	**15 mL/kg over 1 h**, glucose-containing fluid
Shock + severe anaemia	**Blood** (whole blood 20 mL/kg or packed cells 10 mL/kg)
DKA shock	**10–20 mL/kg** 0.9% saline over 20–30 min (ISPAD 2022)
Deficit	**% × kg × 10 mL**
Normal urine output	≥ **1 mL/kg/h** (infants ≥ 2)
Hypoglycaemia	**10% glucose 5 mL/kg** IV
Symptomatic hyponatraemia	**2 mL/kg of 2.7–3% saline over 10–15 min** (max 100 mL), up to 3 doses
Na correction limits	Up: **≤ 8 mmol/L/24 h** (NICE max 12). Down: **≤ 0.5 mmol/L/h** (≤ 10–12/24 h)
Oral KCl	**1–2 mmol/kg/dose** (max 20 mmol/dose; max 5 mmol/kg/day)
IV KCl	**20 mmol/L** in maintenance (max **40 mmol/L** peripheral); acute **0.2 mmol/kg/h** (max 10 mmol/h)
15% KCl	**2 mmol/mL**
Calcium gluconate 10%	**0.5 mL/kg** (max 20 mL; RCH max 30 mL) over 2–5 min
Salbutamol (hyperkalaemia)	**2.5 mg** (≤ 25 kg) / **5 mg** neb
Insulin + glucose	**0.1 unit/kg** + **10% glucose 5 mL/kg**
Sodium bicarbonate 8.4%	**1 mmol/kg (1 mL/kg)** over 30 min, only if acidotic

Item	Value
Resonium	**0.5–1 g/kg** 6-hourly (max 30 g)
Furosemide	**1 mg/kg** IV
Drops/min	20-drop set: **mL/h ÷ 3**; 60-drop micro-drip: **= mL/h**

Exam traps

- **"5% dextrose for maintenance."** Wrong: hypotonic → hyponatraemia; use 0.9% saline (or RL) **with** 5% glucose.
- **"Febrile child with malaria and cold hands → 20 mL/kg bolus."** Wrong unless all 3 signs of shock (FEAST).
- **"Bolus for a child with severe anaemia in shock."** Give **blood**.
- **"Correct Na 116 to 135 today."** Wrong: ≤ 8 mmol/L in 24 h after symptoms settle.
- **"Calcium gluconate lowers potassium."** Wrong: it protects the heart; insulin/glucose and salbutamol lower K.
- **"Add KCl to the first bolus."** Wrong: never in a bolus; only in maintenance once urine flows.
- **"K 6.8 from a heel prick in a well baby → insulin now."** Repeat a free-flowing sample first (if ECG normal).
- **"Fluid-restrict a child with hyponatraemic seizures."** Wrong: give hypertonic saline first.

Questions seniors ask

Viva questions

1. *Calculate maintenance for a 24 kg child.* 1000 + 500 + 80 = 1580 mL/day (≈ 66 mL/h); UCG variant 1600 mL/day.
2. *Why isotonic maintenance fluid?* Sick children have high ADH; hypotonic fluids cause hyponatraemia and encephalopathy (NICE NG29).
3. *What did FEAST show?* Boluses (20–40 mL/kg) of saline or albumin increased 48-h mortality in African children with severe febrile illness and impaired perfusion; WHO now restricts boluses to children with all three signs of shock.
4. *Name three causes of hyperkalaemia in Ugandan children.* AKI (AGN, HUS, malaria), tumour lysis in Burkitt lymphoma, haemolysed samples; also CAH, transfusion.
5. *First drug for hyperkalaemia with peaked T waves?* IV calcium gluconate 10% 0.5 mL/kg over 2–5 min; it stabilises the myocardium but does not lower K.
6. *How fast may you correct hypernatraemia?* No more than 0.5 mmol/L/h, 10–12 mmol/L per 24 h, over 48 h or more.
7. *What is the maximum potassium concentration in a peripheral IV?* 40 mmol/L; maximum rate 0.2 mmol/kg/h on a ward.

On the ward round

- **Write maintenance for this 18 kg child with pneumonia.** Holliday–Segar 1000 + 400 = 1400 mL/day; pneumonia (SIADH risk) → 2/3 ≈ 930 mL/day ≈ 39 mL/h of 0.9% saline with 5% glucose, KCl 20 mmol/L once passing urine; weigh daily, Na/K/glucose at 24 h; switch to oral feeds as soon as possible.
- **Why not bolus every child with a fast pulse and cool hands?** FEAST (East Africa, 2011) showed increased mortality; only children with all three signs of shock get boluses, and SAM, severe anaemia and heart failure have special rules.
- **This child on IV fluids is drowsy. What do you check first?** Glucose (bedside), then sodium; look at the fluid chart (type and volume); look for meningitis or raised ICP.
- **How do you know the child is overloaded?** Weight gain, puffy eyelids, oedema, enlarging tender liver, gallop, crackles, rising RR and falling SpO_2, raised JVP in older children.

- **The K is 7.2 with peaked T waves. What do you do in the next 10 minutes?** Stop all K, monitor, calcium gluconate 10% 0.5 mL/kg IV over 2–5 min, salbutamol neb, insulin 0.1 unit/kg with 10% glucose 5 mL/kg, bicarbonate if acidotic, repeat K and ECG, then Resonium, furosemide if passing urine, and call for dialysis if refractory.
- **How would you make 10% glucose if only 50% and 5% are available?** One part 50% to 9 parts 5% (e.g. 10 mL of 50% + 90 mL of 5%), or one part 50% to 4 parts sterile water.

Mnemonics

- **"4-2-1"** for hourly maintenance; **"100-50-20"** per day.
- **"Deficit = % × kg × 10"**.
- **"No pee, no K"**; **"K max 40 (mmol/L) and 0.2 (mmol/kg/h)"**.
- **"Sodium: slow in, slow out"** (≤ 8–12 up, ≤ 10–12 down per 24 h; 0.5 mmol/L/h).
- **Hyperkalaemia "C BIG K Drop"**: **C**alcium, **B**icarbonate (if acidotic), **I**nsulin + **G**lucose, **K**ayexalate/Resonium, **D**iuretic/**D**ialysis (and salbutamol nebs).
- **"Shock gets a bolus; sick gets 2/3; SAM gets 15; anaemia gets blood."**
- **Drops/min with a 20-drop set = mL/h ÷ 3**; with a 60-drop micro-drip = **mL/h**.

OSCE tips

- **"Write an IV fluid prescription" station**: state the **weight**, show the **calculation**, name the **fluid with glucose**, add **KCl only when passing urine**, give **mL/h and drops/min** for the giving set, and write the **monitoring** (weight, chart, Na/K/glucose at 24 h).
- **"Interpret these electrolytes" station**: **comment on the volume state first**, then sodium, potassium, bicarbonate, urea/creatinine, glucose, and link each to the clinical story.
- **"Assess for fluid overload" station**: weight trend, eyelids, JVP, gallop, crackles, liver edge and span, sacral and pedal oedema, in Hutchison order.

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SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Holliday–Segar maintenance per day	**100 mL/kg** for the first 10 kg + **50 mL/kg** for the next 10 kg + **20 mL/kg** for each kg above 20 kg (UCG 2023 uses 25 mL/kg above 20 kg).
2	Holliday–Segar per hour ('4-2-1')	**4 mL/kg/h** for the first 10 kg, **2 mL/kg/h** for the next 10 kg, **1 mL/kg/h** above 20 kg.
3	Maintenance for a 15 kg child	1000 + 250 = **1250 mL/day ≈ 52 mL/h**.
4	Which fluid for IV maintenance in children over 1 month (NICE NG29)?	**Isotonic** crystalloid (Na 131–154 mmol/L) **with glucose**, e.g. **0.9% saline + 5% glucose** (± KCl 20 mmol/L). Not plain 5% dextrose.
5	Why do sick children become hyponatraemic on IV fluids?	**Non-osmotic ADH release** (pain, nausea, pneumonia, meningitis, surgery) retains free water; **hypotonic fluid** supplies it.
6	Which children get restricted (2/3) maintenance?	High-ADH states: **meningitis, pneumonia, bronchiolitis, head injury, post-op, encephalitis** (NICE: 50–80%). AGN/AKI: insensible losses + urine output.
7	Definition of shock (WHO ETAT 2016)	**All three**: cold extremities, **capillary refill > 3 s**, and a **weak, fast pulse**.
8	Shock bolus volumes: UCG vs WHO ETAT vs NICE	UCG 2023: **20 mL/kg RL/NS as rapidly as possible**. WHO ETAT 2016: **10–20 mL/kg over 30–60 min**. NICE NG29: **10 mL/kg over < 10 min**. Reassess after each.
9	What did the FEAST trial show?	In African children (largely Uganda) with severe febrile illness and **impaired perfusion but not hypotension**, fluid **boluses increased mortality**. No boluses unless full shock.
10	Shock in SAM: fluid and rate	**15 mL/kg over 1 hour** of RL with 5% glucose (or half-strength Darrow's with 5% glucose or 0.45% saline with 5% glucose); repeat once if improving; if not, blood 10 mL/kg over 3 h.
11	Deficit formula	**Deficit (mL) = % dehydration × weight (kg) × 10.** E.g. 7% of 12 kg = 840 mL.
12	Neonatal fluid volumes (UCG 2023)	Day 1 **60**, day 2 **90**, day 3 **120**, day 4+ **150 mL/kg/day** (up to 180 once fully fed); 10% glucose first 2 days.
13	Composition of Ringer's lactate (mmol/L)	Na **130–131**, K **4–5**, Cl **109–111**, lactate **28–29**, Ca **2**; no glucose.
14	Strength of 15% KCl ampoule and how to make 20 mmol/L	**2 mmol/mL.** Add **10 mL to 1 L** (or 5 mL to 500 mL) = 20 mmol/L; mix well.
15	Potassium safety limits on a general ward	Max **40 mmol/L** peripherally; acute IV replacement **0.2 mmol/kg/h (max 10 mmol/h)**; only if **passing urine**; **never IV push or IM**.
16	Oral potassium replacement dose (RCH)	**KCl 1–2 mmol/kg/dose (max 20 mmol/dose)**; max **5 mmol/kg/day** (max 50 mmol/day). Oral route preferred.
17	Emergency treatment of hyponatraemic seizures (NICE NG29)	**2 mL/kg of 2.7% (or 3%) saline over 10–15 min (max 100 mL)**; repeat once; consider a third dose after checking Na.
18	Maximum rise of sodium in 24 h when correcting hyponatraemia	**≤ 8 mmol/L/24 h** (Medscape, RCH 6–8); **NICE: never > 12 mmol/L/24 h**. Risk: osmotic demyelination.
19	Maximum rate of fall of sodium in hypernatraemia	**≤ 0.5 mmol/L per hour** (≈ 10–12 mmol/L per 24 h); replace over **48 h or more**. Risk: cerebral oedema and seizures.

#	Question	Answer
20	First drug in hyperkalaemia with ECG changes	**Calcium gluconate 10% 0.5 mL/kg IV (max 20 mL) over 2–5 min**: stabilises the myocardium (does not lower K).
21	Drugs that shift potassium into cells	**Salbutamol neb** 2.5 mg (≤ 25 kg) / 5 mg; **insulin 0.1 unit/kg + 10% glucose 5 mL/kg**; **sodium bicarbonate 1 mmol/kg** if acidotic.
22	ECG of hyperkalaemia vs hypokalaemia	**High K**: peaked T, prolonged PR, flat/absent P, wide QRS, sine wave, VF/asystole. **Low K**: flat/inverted T, **U waves**, ST depression, long QT, ectopics.
23	Causes of hyperkalaemia in Ugandan children	**AKI** (AGN, HUS, severe malaria), **tumour lysis (Burkitt lymphoma)**, **haemolysed heel-prick sample**, haemolysis, salt-wasting **CAH**, transfusion, ACE inhibitors.
24	Hyponatraemia + hyperkalaemia + hypoglycaemia in a vomiting infant	**Salt-wasting congenital adrenal hyperplasia / adrenal crisis**: saline, **hydrocortisone**, glucose; look for ambiguous genitalia in girls.
25	Making 0.9% saline with 5% glucose on the ward	Remove **50 mL** from a **500 mL** bag of 0.9% saline and add **50 mL of 50% dextrose**; label the bag.
26	Drops per minute formula	**Drops/min = volume (mL) × drop factor ÷ time (min).** 20-drop set: mL/h ÷ 3. 60-drop micro-drip: drops/min = mL/h.
27	Signs of fluid overload in an infant	**Enlarging tender liver**, puffy eyelids, oedema, gallop rhythm, basal crackles, rising RR/falling SpO_2, weight gain > 5%.
28	Monitoring for every child on IV fluids	**Prescription in mL/h**, hourly **fluid chart**, **daily weight**, **Na/K/glucose before and at least every 24 h** (4–6 h if sick), clinical review 4-hourly, switch to oral early.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-year-old (14 kg) admitted with bacterial meningitis cannot feed. She is alert, CRT 2 s, normal pulses. Serum sodium is 129 mmol/L and she is not dehydrated. Which IV fluid prescription is MOST appropriate for the first 24 hours?

A 5% dextrose at 50 mL/h
B 0.9% saline with 5% glucose at about 33 mL/h
C 0.9% saline at 20 mL/kg over 30 minutes, then maintenance
D Half-strength Darrow's at 1200 mL/day
E 0.9% saline with 5% glucose plus KCl 40 mmol/L at 100 mL/h

2. A 4-year-old (16 kg) with severe malaria has fever 39.5 °C, HR 165, cool hands, CRT 2 s and a strong radial pulse. Hb 9 g/dL, BCS 4/5. The ward nurse asks whether to give a 20 mL/kg saline bolus. What is the BEST response?

A Yes, give 320 mL as fast as possible because he is tachycardic
B Give 10 mL/kg albumin instead of saline
C No bolus; start maintenance fluid and reassess within 1 hour
D Give whole blood 20 mL/kg
E Restrict all fluids because of cerebral malaria

3. A 2-year-old (11 kg) with pneumonia, on 5% dextrose 1 L every 12 hours for 2 days, has a generalised seizure. RBS 6.8 mmol/L. Na 119 mmol/L. After securing the airway and giving diazepam, what is the MOST appropriate next step?

A Fluid restriction to half maintenance only
B Furosemide 1 mg/kg IV
C Sodium bicarbonate 1 mmol/kg IV
D Correct the sodium to 135 mmol/L over the next 6 hours with 0.9% saline
E 3% saline 22 mL (2 mL/kg) IV over 10–15 minutes, then recheck sodium

4. A 9-month-old (7.5 kg) with diarrhoea is extremely irritable with doughy skin. Na 164 mmol/L, K 3.6. He is not shocked but cannot drink well. Which plan is MOST appropriate? Advanced

A NG low-osmolarity ORS or IV 0.9% saline with glucose, replacing the deficit over at least 48 h with Na checks every 2–4 h, aiming to lower Na by no more than 0.5 mmol/L/h
B 5% dextrose at 150 mL/kg/day to bring sodium down quickly
C Water by bottle as much as he wants
D 0.45% saline 20 mL/kg bolus
E Restrict fluids to 2/3 maintenance

5. A 6-year-old (20 kg) with Burkitt lymphoma started chemotherapy yesterday. He is oliguric. K 7.4 mmol/L; ECG shows peaked T waves and a widened QRS. What is the FIRST drug to give?

A Calcium polystyrene sulfonate 20 g rectally
B Salbutamol 2.5 mg nebulised
C Calcium gluconate 10% 10 mL IV over 2–5 minutes
D Furosemide 20 mg IV
E Sodium bicarbonate 8.4% 20 mL IV

6. A 5-year-old (18 kg) with diarrhoea has K 2.6 mmol/L. He is alert, drinking, passing urine, with no ECG changes. What is the MOST appropriate potassium treatment?

A KCl 15% 5 mL IV push
B IV KCl 60 mmol in 500 mL through a peripheral cannula over 2 hours
C No treatment; it will correct itself
D KCl 1–2 mmol/kg/dose orally (max 20 mmol per dose), recheck K, and check magnesium
E Insulin and glucose

7. A 12 kg child is 5% dehydrated (not shocked), cannot take orally after abdominal surgery, and you prescribe deficit plus maintenance over 24 hours. Using a 20 drops/mL giving set, what is the approximate drip rate?

A About 14 drops/min
B About 47 drops/min
C About 70 drops/min
D About 25 drops/min
E About 100 drops/min

8. A 3-week-old girl presents with vomiting and poor feeding. She is lethargic. Na 124, K 7.1, glucose 2.4 mmol/L. On examination there is clitoromegaly with fused labia. What is the MOST likely diagnosis? Advanced

A Salt-wasting congenital adrenal hyperplasia
B Pyloric stenosis
C SIADH from meningitis
D Acute gastroenteritis with dehydration
E Nephrotic syndrome

9. A 2-year-old (10 kg) with Hb 3.8 g/dL from severe malaria has cold hands, CRT 4 s and a weak, fast pulse. What fluid is MOST appropriate?

A 0.9% saline 20 mL/kg rapidly, then transfuse later
B 5% dextrose 10 mL/kg
C Packed red cells or whole blood as soon as possible, with other IV fluid only to maintain hydration
D Ringer's lactate 40 mL/kg over 1 hour
E No fluids until the Hb is rechecked

10. Which of the following is the LEAST appropriate statement about potassium chloride (KCl) for IV use in children?

A 15% KCl contains 2 mmol/mL
B Add potassium only once the child is passing urine
C Mix the bag thoroughly after adding KCl
D On a general ward the maximum acute IV rate is about 0.2 mmol/kg/h
E KCl can be given IM if IV access fails

11. A 7-year-old (22 kg) with acute post-streptococcal glomerulonephritis has periorbital oedema, BP 140/95 and passed 200 mL of urine in the last 24 hours. Creatinine 180 µmol/L, K 5.6. Which fluid plan is MOST appropriate? Advanced

A Full Holliday–Segar maintenance IV
B Insensible losses (about 400 mL/m²/day) plus the previous day's urine output, with salt restriction and no added potassium
C 20 mL/kg 0.9% saline bolus to improve urine output
D Ringer's lactate at 2/3 maintenance
E Unrestricted oral fluids because he is thirsty

12. You need 10% dextrose for a hypoglycaemic child but the ward only has 50% dextrose and 5% dextrose. How do you prepare about 100 mL of approximately 10% dextrose?

- A 50 mL of 50% dextrose + 50 mL of 5% dextrose
- B 20 mL of 50% dextrose + 80 mL of 5% dextrose
- C Give 50% dextrose undiluted at 1 mL/kg
- D 10 mL of 50% dextrose + 90 mL of 5% dextrose
- E 5 mL of 50% dextrose + 95 mL of sterile water

13. A 4-year-old (16 kg) with severe acute malnutrition (MUAC 10.9 cm, oedema +) is lethargic with cold hands, CRT 4 s and a weak, fast pulse. Which fluid order is MOST appropriate? Advanced

- A Ringer's lactate 320 mL as fast as possible, repeat if needed
- B Plan C: 100 mL/kg over 3 hours
- C Ringer's lactate with 5% glucose 240 mL over 1 hour, then reassess
- D ReSoMal 80 mL every 30 minutes orally
- E 0.9% saline 160 mL over 10 minutes

14. A well 5-day-old baby has a routine heel-prick sample showing K 6.9 mmol/L. She is feeding well, passing urine, and her ECG is normal. What is the MOST appropriate next step?

- A Give calcium gluconate 10% immediately
- B Start insulin and glucose
- C Give Resonium rectally
- D Repeat potassium on a free-flowing venous sample
- E Arrange peritoneal dialysis

15. A 30 kg child is nil by mouth after appendicectomy. Using Holliday–Segar, what is the full daily maintenance volume?

- A 1500 mL
- B 1700 mL
- C 2000 mL
- D 3000 mL
- E 1200 mL

Answers & explanations

> **1. B** · 0.9% saline with 5% glucose at about 33 mL/h
>
> B is correct: maintenance = 1000 + 200 = 1200 mL/day (50 mL/h); meningitis raises ADH and her sodium is already low (SIADH), so give about 2/3 (≈ 800 mL/day ≈ 33 mL/h) of an isotonic fluid with glucose (NICE NG29). (Without hyponatraemia or raised ICP, WHO and Cochrane 2016 favour full isotonic maintenance in meningitis.) A is wrong: plain 5% dextrose is hypotonic and, at full maintenance, risks hyponatraemic seizures. C is wrong: she is not shocked, so no bolus. D is wrong: half-strength Darrow's is hypotonic and the volume is unrestricted. E is wrong: 100 mL/h is twice maintenance, and potassium should not start until urine output is confirmed.

2. C · No bolus; start maintenance fluid and reassess within 1 hour

C is correct: he has only one sign of impaired circulation (cool hands) and a fast pulse from fever, not shock (which needs cold extremities plus CRT > 3 s plus a weak fast pulse). WHO ETAT 2016, after FEAST, advises maintenance fluids and reassessment within an hour. A is wrong: boluses increased mortality in febrile African children without shock (FEAST). B is wrong: albumin boluses also increased mortality in FEAST. D is wrong: Hb 9 g/dL does not need transfusion. E is wrong: he still needs maintenance (glucose-containing) fluid; complete restriction risks hypoglycaemia and dehydration.

3. E · 3% saline 22 mL (2 mL/kg) IV over 10–15 minutes, then recheck sodium

E is correct: acute symptomatic hyponatraemia (seizure, Na 119) needs hypertonic saline 2 mL/kg (max 100 mL) over 10–15 min, repeated if symptoms persist (NICE NG29); stop the hypotonic fluid. A is wrong: NICE states hyponatraemic encephalopathy must not be managed by fluid restriction alone. B is wrong: furosemide alone does not correct encephalopathy and may worsen hypovolaemia. C is wrong: bicarbonate has no role. D is wrong: raising Na by 16 mmol/L in 6 h risks osmotic demyelination; after symptoms stop, aim for ≤ 8 mmol/L (NICE max 12) in 24 h.

4. A · NG low-osmolarity ORS or IV 0.9% saline with glucose, replacing the deficit over at least 48 h with Na checks every 2–4 h, aiming to lower Na by no more than 0.5 mmol/L/h

A is correct: hypernatraemic dehydration is corrected slowly (≤ 0.5 mmol/L/h, about 10–12 mmol/L/24 h) over ≥ 48 h with isotonic fluid or ORS (NICE allows ORS in hypernatraemia) and frequent sodium checks. B is wrong: a large hypotonic load drops sodium fast and causes cerebral oedema and seizures. C is wrong: free water alone also corrects too fast. D is wrong: boluses are for shock and should be isotonic. E is wrong: he is dehydrated and needs deficit replacement, not restriction.

5. C · Calcium gluconate 10% 10 mL IV over 2–5 minutes

C is correct: with ECG changes, the first priority is to protect the myocardium with IV calcium gluconate 10% 0.5 mL/kg (10 mL) over 2–5 minutes; it acts within 3 minutes. Then shift K (salbutamol, insulin + glucose) and remove it. A is wrong: resins act over hours. B is wrong: salbutamol is the right second step but takes 30 minutes and does not protect the heart. D is wrong: he is oliguric (tumour lysis AKI), so furosemide will do little, and it is not first-line. E is wrong: bicarbonate is only for acidosis and should not be first; never give it with calcium in the same line.

6. D · KCl 1–2 mmol/kg/dose orally (max 20 mmol per dose), recheck K, and check magnesium

D is correct: the oral route is preferred when the child can drink and there are no ECG changes (RCH): 1–2 mmol/kg/dose (max 20 mmol), repeat after checking K, max 5 mmol/kg/day; check Mg. A is wrong: KCl IV push can cause cardiac arrest; it must always be diluted. B is wrong: 120 mmol/L exceeds the 40 mmol/L peripheral limit and the rate (30 mmol/h) exceeds 0.2 mmol/kg/h (3.6 mmol/h; max 10 mmol/h). C is wrong: K 2.6 is moderate-severe and needs treatment. E is wrong: insulin lowers potassium further.

7. D · About 25 drops/min

D is correct: deficit = 5 × 12 × 10 = 600 mL; maintenance = 1000 + 100 = 1100 mL; total 1700 mL/24 h ≈ 71 mL/h; with a 20-drop set, drops/min = 71 × 20 ÷ 60 ≈ 24. (In practice, as a post-op child with high ADH, you would restrict maintenance and check electrolytes.) A is wrong: about 14–15 drops/min (≈ 45 mL/h) replaces maintenance only and ignores the 600 mL deficit. B is wrong: 47 drops/min would be about 140 mL/h, double the requirement. C is wrong: 70 drops/min is the mL/h figure, which equals drops/min only with a 60-drop micro-drip set. E is wrong: 100 drops/min would give 300 mL/h, a dangerous overload.

8. A · Salt-wasting congenital adrenal hyperplasia

A is correct: hyponatraemia with hyperkalaemia and hypoglycaemia in a 2–3-week-old with virilised genitalia is classic salt-wasting CAH (21-hydroxylase deficiency): give 0.9% saline, hydrocortisone and glucose, and refer. B is wrong: pyloric stenosis gives hypochloraemic hypokalaemic alkalosis, not hyperkalaemia. C is wrong: SIADH causes euvolaemic hyponatraemia with normal potassium. D is wrong: diarrhoea usually causes low potassium, and it does not explain the genitalia. E is wrong: nephrotic syndrome presents with oedema and proteinuria, rarely in neonates, without hyperkalaemia.

9. C · Packed red cells or whole blood as soon as possible, with other IV fluid only to maintain hydration

C is correct: WHO ETAT 2016 recommends that children in shock with severe anaemia (Hb < 5 g/dL) receive blood as early as possible (whole blood 20 mL/kg or packed cells 10 mL/kg) and other fluids only for maintenance. A is wrong: crystalloid boluses dilute an already critical Hb and can precipitate heart failure. B is wrong: dextrose is never a resuscitation fluid. D is wrong: a 40 mL/kg crystalloid load is dangerous in severe anaemia. E is wrong: delaying treatment in shock is fatal; the Hb is already known.

10. E · KCl can be given IM if IV access fails

E is the answer (the least appropriate, and dangerous statement): KCl must never be given IM or as an IV push; it causes tissue necrosis and cardiac arrest. Statement A is true: 15% KCl = 150 mg/mL = 2 mmol/mL. Statement B is true: 'no pee, no K', because potassium accumulates in oliguria. Statement C is true: KCl layers at the bottom of an unmixed bag, giving a concentrated bolus. Statement D is true: RCH limits acute replacement on a general ward to 0.2 mmol/kg/h (max 10 mmol/h).

11. B · Insensible losses (about 400 mL/m²/day) plus the previous day's urine output, with salt restriction and no added potassium

B is correct: in AGN with oliguria and fluid overload, give insensible losses plus urine output, restrict salt, avoid potassium, and use furosemide and antihypertensives as needed. A is wrong: full maintenance (1540 mL/day) would worsen overload and hypertension. C is wrong: a bolus in an overloaded child risks pulmonary oedema and hypertensive encephalopathy. D is wrong: RL contains potassium and 2/3 maintenance is still too much. E is wrong: unrestricted fluids worsen overload.

12. D · 10 mL of 50% dextrose + 90 mL of 5% dextrose

D is correct: 10 mL of 50% (5 g) + 90 mL of 5% (4.5 g) = 9.5 g in 100 mL ≈ 10% (UCG: 1 part 50% with 9 parts 5%). A is wrong: that gives about 27.5%, hyperosmolar and irritant to small veins. B is wrong: that gives 14%. C is wrong: 50% dextrose is sclerosing and causes rebound; dilute it. E is wrong: that gives only 2.5%.

13. C · Ringer's lactate with 5% glucose 240 mL over 1 hour, then reassess

C is correct: shock in SAM = 15 mL/kg (240 mL) over 1 hour of a glucose-containing fluid (RL with 5% glucose, half-strength Darrow's with 5% glucose or 0.45% saline with 5% glucose); repeat once if improving, then oral/NG ReSoMal; if not improving, blood 10 mL/kg over 3 h (UCG 2023; WHO ETAT 2016). A is wrong: 20 mL/kg rapid boluses risk heart failure in SAM. B is wrong: Plan C is for well-nourished children. D is wrong: oral ReSoMal is for SAM without shock. E is wrong: a 10 mL/kg rapid bolus (NICE practice) is not the SAM protocol and lacks glucose.

14. D · Repeat potassium on a free-flowing venous sample

D is correct: squeezed heel-prick samples are often haemolysed, falsely raising potassium; in a well baby with a normal ECG, repeat on a free-flowing venous sample first (neonatal K also runs slightly higher). A is wrong: calcium is for ECG changes or severe confirmed hyperkalaemia. B is wrong: insulin risks hypoglycaemia in a newborn for a probably spurious result. C is wrong: resins in neonates can cause bowel injury and are not needed here. E is wrong: dialysis is for refractory, confirmed hyperkalaemia with AKI.

15. B · 1700 mL

B is correct: 1000 (first 10 kg) + 500 (next 10 kg) + 200 (10 kg × 20) = 1700 mL/day (≈ 70 mL/h); after surgery many children need less (NICE 50–80%). A is wrong: 1500 mL is the value for 20 kg. C is wrong: 2000 mL forgets that above 20 kg the rate falls to 20 mL/kg (UCG's 25 mL/kg variant gives 1750). D is wrong: 3000 mL uses 100 mL/kg for every kilogram. E is wrong: 1200 mL is roughly 2/3 of maintenance, not the full volume.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Seizure on IV fluids: iatrogenic hyponatraemia

You are the intern on the paediatric ward at a regional referral hospital. Musa, a 2-year-old boy (12 kg), was admitted yesterday with severe pneumonia. He has been on '5% dextrose 1 L 12-hourly' since admission and is not feeding. At 06:00 the nurse calls: he has had a generalised seizure lasting 5 minutes.

1. What are your immediate actions?

> **ABC**: position, suction, **oxygen** (SpO_2 86% → nasal prongs). If still seizing: **diazepam 0.5 mg/kg rectally** (or 0.3 mg/kg IV slowly). **Check glucose** at the bedside (6.5 mmol/L). **Stop the 5% dextrose.** Send urgent **electrolytes** and creatinine. Call the senior.

2. How much fluid has he received compared with what he needed?

> He received **2 L/day of plain 5% dextrose**. Maintenance for 12 kg = 1000 + 100 = **1100 mL/day**, and with pneumonia (SIADH risk) about **2/3 = 730 mL/day**. He got nearly **3 times** the appropriate volume of a **hypotonic** fluid.

3. Na comes back at 118 mmol/L. He is drowsy and twitching. What do you give?

> **Hypertonic saline 2 mL/kg = 24 mL of 3% saline (or 2.7%) IV over 10–15 min** (NICE NG29; max 100 mL). Repeat once if symptoms persist; recheck Na and consider a third dose. Expect a rise of about 4–6 mmol/L to stop the seizures.

4. The twitching stops. Na at 1 hour is 123. Write the ongoing fluid prescription and monitoring.

> - **0.9% saline with 5% glucose** at **2/3 maintenance ≈ 30 mL/h**; KCl 20 mmol/L once urine output is confirmed
> - **Na every 2–4 h**: total rise **≤ 8 mmol/L in 24 h** after symptoms stop (NICE max 12): target ≤ 126 at 24 h
> - Hourly **fluid chart**, **weigh** now and in 12 h, glucose 4-hourly
> - Start **NG feeds** (expressed breast milk/F-75 not needed; normal milk feeds) as soon as safe and reduce IV volume accordingly

5. Six hours later he is passing large volumes of dilute urine and Na is 128 (up 10 in 7 h). What is happening and what do you do?

> As ADH switches off he has a **water diuresis**, and sodium is rising **too fast** (risk of osmotic demyelination). Inform the senior; **slow or stop the saline**, consider **5% glucose** or oral water to slow the rise, and recheck Na in 1–2 h.

6. What system changes would you propose at the morning meeting?

> Remove plain 5% dextrose as a default maintenance fluid; ward chart for **mL/h by weight** (4-2-1) and **2/3 rule** for meningitis and pneumonia; ready-made **0.9% saline + 5% glucose** recipe on the wall; **electrolytes at 24 h** for every child on IV fluids; burettes or micro-drip sets for children under 15 kg; report the incident without blame.

Take-away: Plain 5% dextrose plus high ADH equals hyponatraemic seizures: use isotonic fluid with glucose at 2/3 maintenance, treat seizures with 2 mL/kg hypertonic saline, and then correct slowly.

Case 2: Hyperkalaemia in a child with acute kidney injury

Grace, a 6-year-old girl (18 kg), is admitted to Mengo Hospital with 3 days of fever, vomiting and dark ('Coca-Cola') urine after treatment for malaria. She has passed very little urine today. She is drowsy but rousable.

1. What is your initial assessment and what do you send?

> **ABC, ETAT**: RR 36, SpO_2 95%, HR 120, BP 118/76, CRT 2 s, not shocked; **pallor**, mild jaundice, no oedema; BCS 4/5. Bladder not palpable. Bedside **RBS** (5.2), **HemoCue** (Hb 6.4), malaria RDT/smear, **urine dipstick** (blood +++). Send **electrolytes, urea, creatinine**, FBC, group and cross-match. Insert a urinary catheter to measure output. Likely **blackwater fever / severe malaria with AKI**.

2. Results: K 7.0 mmol/L, Na 131, urea 28, creatinine 340 µmol/L, bicarbonate 14. ECG shows peaked T waves. Write your immediate orders.

> 1. **Stop all potassium** (no RL); cardiac monitor
> 2. **Calcium gluconate 10% 9 mL (0.5 mL/kg) IV over 2–5 min**; repeat if ECG still abnormal
> 3. **Salbutamol 2.5 mg nebulised** (≤ 25 kg), repeat 1–2-hourly
> 4. **Soluble insulin 1.8 units (0.1 unit/kg) IV + 10% glucose 90 mL (5 mL/kg)**; glucose every 30–60 min
> 5. **Sodium bicarbonate 8.4% 18 mL (1 mmol/kg) over 30 min** (she is acidotic), not in the calcium line
> 6. Repeat **K and ECG** in 1–2 h
> 7. Resonium 0.5–1 g/kg 6-hourly orally if K remains high

3. How will you manage her fluids?

> She is not shocked and is oliguric: give **insensible losses (about 400 mL/m²/day; BSA ≈ 0.75 m² → 300 mL/day) plus the previous hour's/day's urine output**, as **potassium-free** fluid with glucose (0.9% saline with 5% glucose); weigh 12-hourly; strict fluid chart. Consider **furosemide 1–2 mg/kg IV** once to test for a response. Transfuse cautiously for Hb 6.4 if symptomatic (10 mL/kg packed cells over 3–4 h with monitoring).

4. Which features would make you call for dialysis?

> **Refractory hyperkalaemia**, **severe acidosis**, **fluid overload with pulmonary oedema** or uncontrolled hypertension, **uraemic encephalopathy**, rising creatinine with **anuria**. Peritoneal dialysis is available at **Mulago** (and some RRHs): arrange early transfer with a referral letter.

5. Continue malaria treatment: anything to adjust?

> **IV artesunate** continues (no renal dose change). Avoid nephrotoxins (NSAIDs, gentamicin unless essential with level/dose adjustment). Review drug doses for AKI. Monitor glucose (malaria and insulin both cause hypoglycaemia).

6. Present her on the ward round.

> 'Grace, 6 years, 18 kg, fully immunised, HIV-negative, with severe malaria complicated by **haemoglobinuria and oliguric AKI** (creatinine 340), **hyperkalaemia 7.0 with ECG changes** treated with calcium, salbutamol, insulin–glucose and bicarbonate, **metabolic acidosis** and **anaemia** (Hb 6.4). On restricted K-free fluids (insensible + output), IV artesunate. Plan: K and ECG 2-hourly, discuss dialysis with Mulago if K or creatinine keep rising.'

Take-away: Hyperkalaemia with ECG changes: protect the heart with calcium, shift potassium with salbutamol and insulin–glucose, remove it, stop all K, restrict fluids to insensible losses plus output, and refer early for dialysis.

Case 3: Hypernatraemic dehydration in a formula-fed infant

Baby Ivan, 4 months old, is brought to the emergency unit with 3 days of diarrhoea and fever. His mother returned to work and he is formula-fed; she has been adding extra scoops 'to make him strong'. He weighs 5.4 kg (6.0 kg at the 14-week immunisation visit).

1. What do you find and how do you interpret it?

> He is **very irritable**, high-pitched cry, **intensely thirsty**, **doughy skin**, eyes slightly sunken, CRT 2 s, warm hands, HR 170, temperature 38.4 °C. Weight loss 0.6 kg = **10% dehydration**, more than the signs suggest. Suspect **hypernatraemic dehydration** (concentrated feeds + fever + diarrhoea).

2. Na 158 mmol/L, K 3.2, glucose 7.0. Is he shocked, and what is your fluid strategy?

> Not shocked (warm hands, CRT 2 s), so **no bolus**. Rehydrate **slowly**: ORS by **NG tube** (or IV 0.9% saline with 5% glucose) replacing the deficit (10% × 5.4 × 10 = 540 mL) plus maintenance (540 mL/day) over **48 h** ≈ 1620 mL/48 h ≈ **34 mL/h**. Aim for a fall in Na of **≤ 0.5 mmol/L/h** (≤ 10–12 mmol/L/24 h).

3. How will you monitor?

> **Na 2–4-hourly** initially (RCH: 1–2 h after starting, then 4–6-hourly), glucose, **neurological checks** (irritability, fontanelle, seizures), weight 12-hourly, fluid chart. Add **KCl 20 mmol/L** to IV fluid once passing urine (K 3.2).

4. At 6 hours Na is 150 (fall of 8 in 6 h). What do you do?

> That is **1.3 mmol/L/h, too fast**. **Slow the infusion** (or change to a more sodium-rich fluid), inform the senior, recheck Na in 1–2 h, and watch for **seizures** (cerebral oedema). If he convulses: stop hypotonic fluid, give **3% saline 2–3 mL/kg**, glucose check, ICU.

5. What else does he need?

> **Zinc 10 mg daily for 10 days** (under 6 months); look for the cause of fever (malaria RDT, urine, ears, chest); continue feeding with **correctly prepared** milk once rehydrating; consider whether he can return to **breastfeeding** (relactation support). Check immunisations (14-week vaccines given).

6. How do you counsel his mother?

> Show **correct formula preparation**: boiled water cooled, **1 level scoop per 30 mL** (or per tin instructions), never extra scoops or sugar; cup feeding rather than a bottle; hygiene. Explore **expressing breast milk** for when she is at work, or relactation. Teach ORS for future diarrhoea and **danger signs**.

Take-away: Hypernatraemia hides dehydration: weigh the baby, rehydrate over 48 h, measure sodium often, slow down if it falls faster than 0.5 mmol/L/h, and fix the feeding error.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.